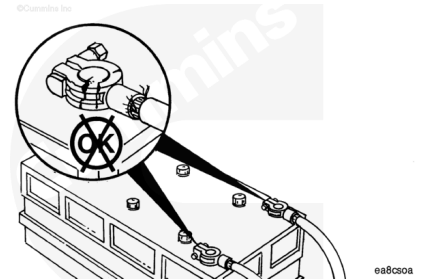


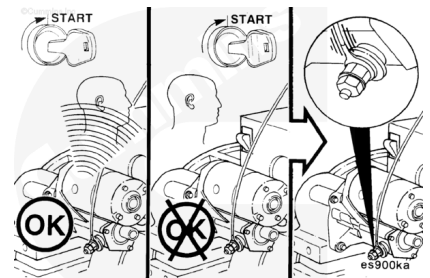
Trainings ▾

Initial Check

Before troubleshooting the starting motor, make sure the battery terminals are **not** loose or corroded.

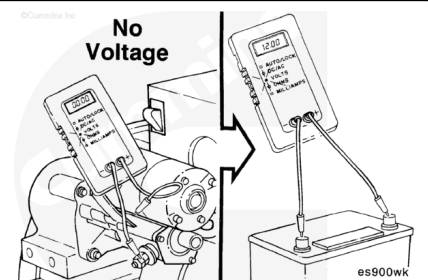


If the starting motor solenoid does **not** make an audible sound, check for loose wiring connections.



Use a digital multimeter, Cummins Part Number 3164488, or equivalent, to set the voltage scale.

Check for system voltage at the starting motor solenoid battery terminal.

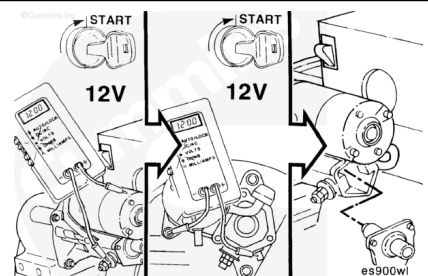


If the multimeter indicates system voltage at the starting motor battery terminal, check the voltage at the starting motor solenoid S terminal, when the starting switch is energized.

If the multimeter indicates system voltage at the S terminal, but the starter does **not** engage, the starting motor solenoid is malfunctioning and the starter **must** be replaced. Refer to

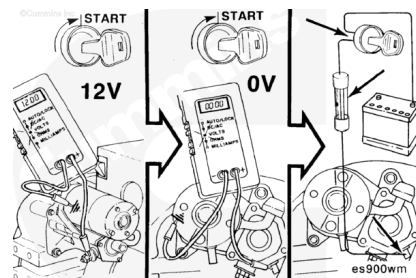
Procedure 013-020

(/qs3/pubsys2/xml/en/procedures/100/100-013-020.html).



If the multimeter does **not** indicate system voltage at the S terminal, check:

- Fuses
- Voltage to the ignition switch and magnetic switch. Refer to "Starting Motor Switch and Magnetic Switch - Checking" in this section. Refer to Procedures 013-017 (</qs3/pubsys2/xml/en/procedures/100/100-013-017.html>) and 013-018 (</qs3/pubsys2/xml/en/procedures/100/100-013-018.html>)
- Application safety shutoff systems

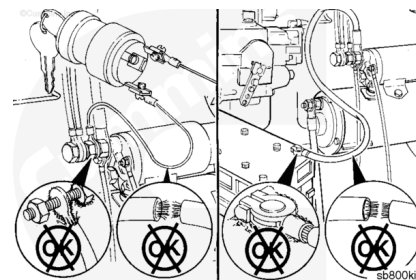


Voltage Check

Set the digital multimeter, Part Number 3164488, to measure DC voltage.

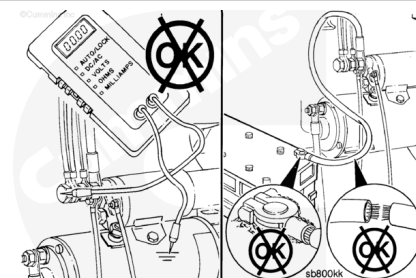
Connect the multimeter positive (+) lead to the starter solenoid positive cable terminal and the negative (-) lead to a chassis or engine ground location.

The multimeter **must** show a voltage with the starter switch "OFF" to be normal.



If the multimeter does **not** indicate a voltage, check the cable connecting the starter solenoid and battery for breaks.

Also check for loose or corroded connections.

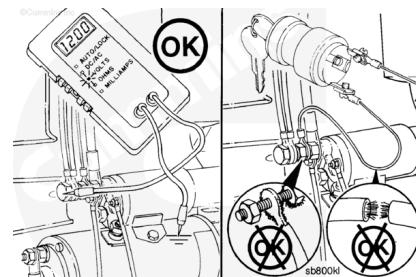


If the multimeter indicates a voltage, but the starter will **not** operate, check the wire connecting the starter solenoid to the starter switch for breaks.

Also check for loose or corroded connections.

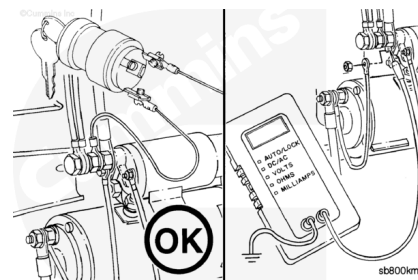
Be sure to check for:

- Fuses
- Application engine shutoff systems.



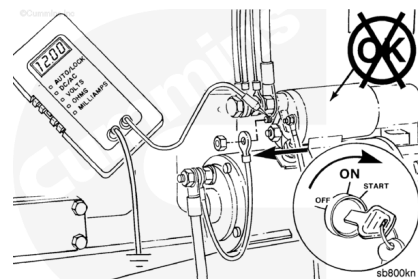
If the wire connecting the starter solenoid and starter switch is **not** loose or damaged, and the starter will **not** operate, remove the cable connecting the starter and starter solenoid from the solenoid terminal.

Connect the multimeter positive (+) lead to the solenoid positive (+) terminal and the negative (-) lead to the chassis or an engine ground location.

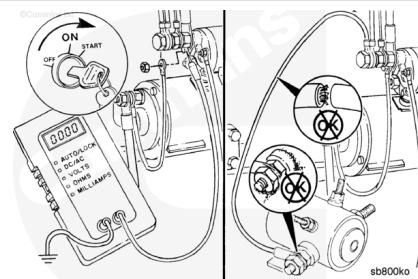


Turn the starter switch to the “START” position.

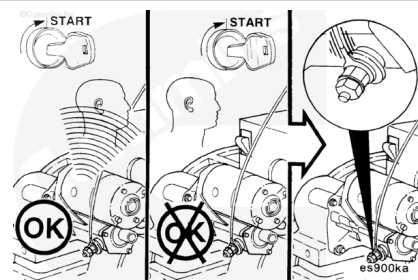
If the multimeter indicates a voltage, the starter solenoid is malfunctioning and **must** be replaced.



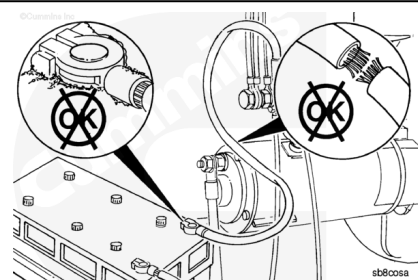
If the multimeter does **not** indicate a voltage, check the wire connecting the starter solenoid to the magnetic switch for breaks, and for loose or corroded connections.



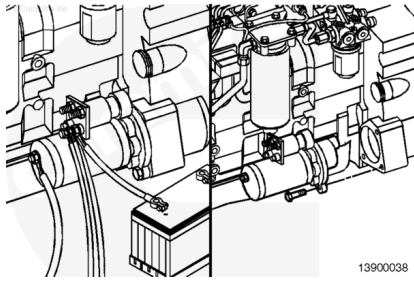
If the wire connecting the starter solenoid to the magnetic switch is **not** loose or damaged and the starter will **not** operate, check the cable connecting the starter solenoid to the starting motor for breaks, and for loose or corroded connections.



Check the cable connecting the starting motor to the battery for breaks, and for loose or corroded connections.



If the cables are **not** loose or damaged, the starting motor is defective and **must** be replaced. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/100/100-013-020.html>).



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Last Modified: 03-Sep-2003
